

Cyclistic Bike-Share Case Study

How do annual members and casual riders use Cyclistic bikes differently?

Google Data Analytics Capstone | 2025 Trip Data

Prepared October 2026

Executive Summary

Business question. This analysis examines how annual members and casual riders use Cyclistic differently so that marketing strategies can be better targeted toward converting casual riders into annual members.

Main result. The strongest pattern is that casual riders behave more like occasional, seasonal, and weekend-oriented users, while annual members use the service more consistently. Casual riders take much longer trips, are more concentrated on weekends, and show a sharper summer peak.

Metric	Annual Members	Casual Riders
Average ride length	12.3 min	22.6 min
Average ride length (seconds)	739.9	1,355.8
Peak ridership month	August	August
Most common riding day	Thursday	Saturday

Recommended direction. Cyclistic should focus conversion campaigns on high-use summer periods and weekend riders, while testing messages that explain the value of membership for riders who already use Cyclistic repeatedly.

1. Business Task

Cyclistic is a fictional bike-share company used in the Google Data Analytics capstone. The marketing team wants to increase annual memberships because annual members are considered more valuable to the business. The assigned question for this case study is: “How do annual members and casual riders use Cyclistic bikes differently?”

The purpose of this analysis was to identify measurable differences in riding behavior between the two groups and translate those differences into marketing recommendations that could help convert casual riders into annual members.

2. Data Sources

I used Cyclistic’s 2025 historical trip data, consisting of 12 monthly datasets covering January through December 2025. The underlying public trip data is from Chicago’s Divvy bicycle-sharing service and is made available by Motivate International Inc. under its [data license agreement](#).

The monthly datasets contain ride-level information such as ride ID, bike type, start and end times, station information, rider membership type, and location fields. For the case study, I also created ride-length and day-of-week fields to support the required descriptive analysis.

The data is appropriate for comparing usage patterns because it records individual trips and whether the rider was an annual member or a casual rider. However, it does not contain personally identifiable information or a stated purpose for each trip, so conclusions about commuting, tourism, or recreation must be treated as interpretations rather than confirmed rider motives.

3. Data Cleaning and Preparation

3.1 Excel preparation

The original CSV files were opened in Excel and saved as Excel workbooks so the monthly files could be reviewed and prepared consistently. I checked each month for formatting problems, duplicate records, missing values, and inconsistent data types.

A `ride_length` column was created by subtracting each ride's start time from its end time and formatting the result as HH:MM:SS. A `day_of_week` column was also created using Excel's WEEKDAY function, with Sunday coded as 1 and Saturday coded as 7.

I confirmed that `ride_id` did not contain duplicate values. Some station-name fields were missing, but those rows were retained because station information was not necessary for the main business question and removing them would have discarded otherwise useful ride behavior data.

3.2 BigQuery preparation

After the Excel cleaning steps, the files were converted back to CSV and imported into BigQuery as 12 monthly tables. The monthly tables were combined into a full-year dataset using UNION ALL. During the combination step, I added `month_number` and `month_name` fields so the full-year data could be analyzed by month.

The `ride_length` field was imported into BigQuery as a string. The monthly source files did not use a completely consistent hour format: some values appeared as 0:15:30 while others appeared as 00:15:30. I used LPAD() to standardize each value to eight characters, then used SUBSTR() and CAST() to convert the hour, minute, and second portions into total seconds.

The derived field, **`ride_length_seconds`**, was stored as a numeric value so functions such as AVG() could be used reliably. The final cleaned dataset was saved as **Cyclistic_2025_Clean**.

3.3 Analysis workflow

BigQuery was used to create summary statistics and analysis tables for rider type, ride duration, day of week, month, and bike type. Those summary results were brought into Google Sheets using Connected Sheets and then connected to Tableau Public for visualization.

4. Analysis

4.1 Bike type by membership

I grouped rides by membership type and calculated the share of each group's rides taken on electric bikes. Casual riders used electric bikes for 72.46% of their rides, compared with 64.71% of rides taken by annual members.

Electric bikes therefore represented a larger share of casual usage. This calculation was kept as a supporting finding even though it was not included in the final Tableau dashboards.

4.2 Average ride length

I calculated average ride length for all riders and then separately for annual members and casual riders. The results were:

- All riders: 961.7 seconds (16.0 minutes)
- Annual members: 739.9 seconds (12.3 minutes)
- Casual riders: 1,355.8 seconds (22.6 minutes)

Casual riders averaged approximately 10.3 minutes longer per ride than annual members. Their average ride was about 83% longer overall. This is one of the clearest behavioral differences between the two groups.

Longer casual rides are consistent with more recreational or leisure-oriented use, while shorter member rides may be more consistent with routine transportation, commuting, or errands. The data does not contain trip-purpose information, so those explanations cannot be confirmed from this dataset alone.

4.3 Day of the week patterns

I counted rides by day of week and separated the totals by membership type. Thursday was the most common riding day for annual members, while Saturday was the most common day for casual riders.

The Tableau visualization shows that member ridership remains relatively high throughout the workweek, especially Tuesday through Thursday. Casual ridership is lowest in the middle of the week and rises toward the weekend, reaching its highest level on Saturday.

I also calculated average ride length for each day of the week. Casual riders had a longer average ride than members on every day of the week. Their longest average rides occurred on Sunday at 1,568.4 seconds (26.1 minutes) and Saturday at 1,515.4 seconds (25.3 minutes). Their shortest average ride occurred on Wednesday at 1,123.0 seconds (18.7 minutes).

Members showed the same general weekend increase in ride duration but at much lower levels. Their longest average rides occurred on Sunday at 818.6 seconds (13.6 minutes) and Saturday at 809.0 seconds (13.5 minutes), while Wednesday was the shortest day at 707.1 seconds (11.8 minutes).

The weekly results therefore show two differences at once: casual riders become more active toward the weekend, and their weekend trips are also substantially longer. Members use Cyclistic more consistently throughout the workweek and generally take shorter trips.

4.4 Monthly ridership

I counted rides by month and separated the results between annual members and casual riders. Ridership for both groups increased during the warmer months, and both groups reached their highest monthly ridership in August.

Annual members completed more rides than casual riders in every month of 2025. Their ridership also remained much more stable as summer ended. From August to September, member ridership decreased by only 0.69%, while casual ridership decreased by 21.4%.

The sharper decline among casual riders indicates that casual usage is more seasonal. Their ridership rises rapidly during spring and summer and then drops much faster at the end of the peak season. Member ridership also changes with the seasons, but annual members continue to use the system at relatively high levels throughout the year.

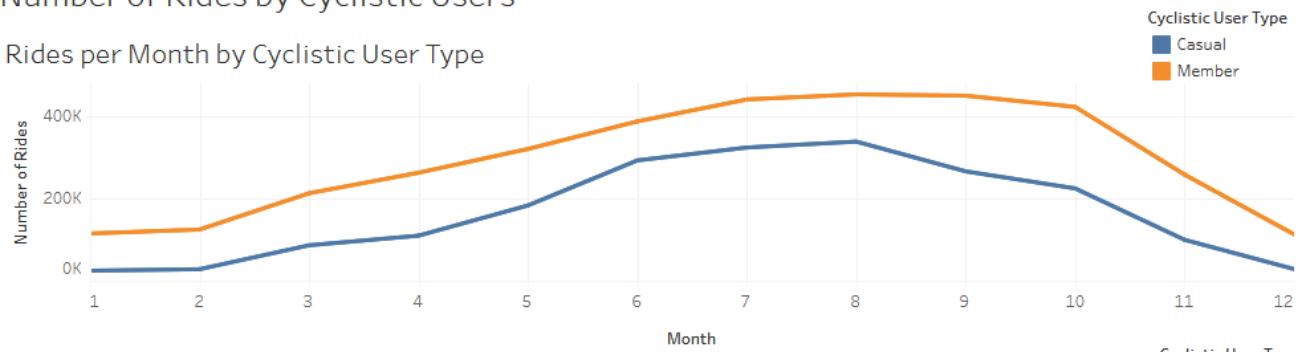
Weather, tourism, school calendars, and recreation are possible explanations for the seasonal difference, but additional data would be needed to determine which factors are responsible.

5. Supporting Visualizations and Key Findings

Two Tableau Public dashboards were created to communicate the main findings. The first focuses on ride frequency by month and day of week. The second focuses on average ride duration overall and by day of week. Together, the dashboards show both when the two groups ride and how long they ride.

Number of Rides by Cyclistic Users

Rides per Month by Cyclistic User Type



Rides per Day by Cyclistic User Type

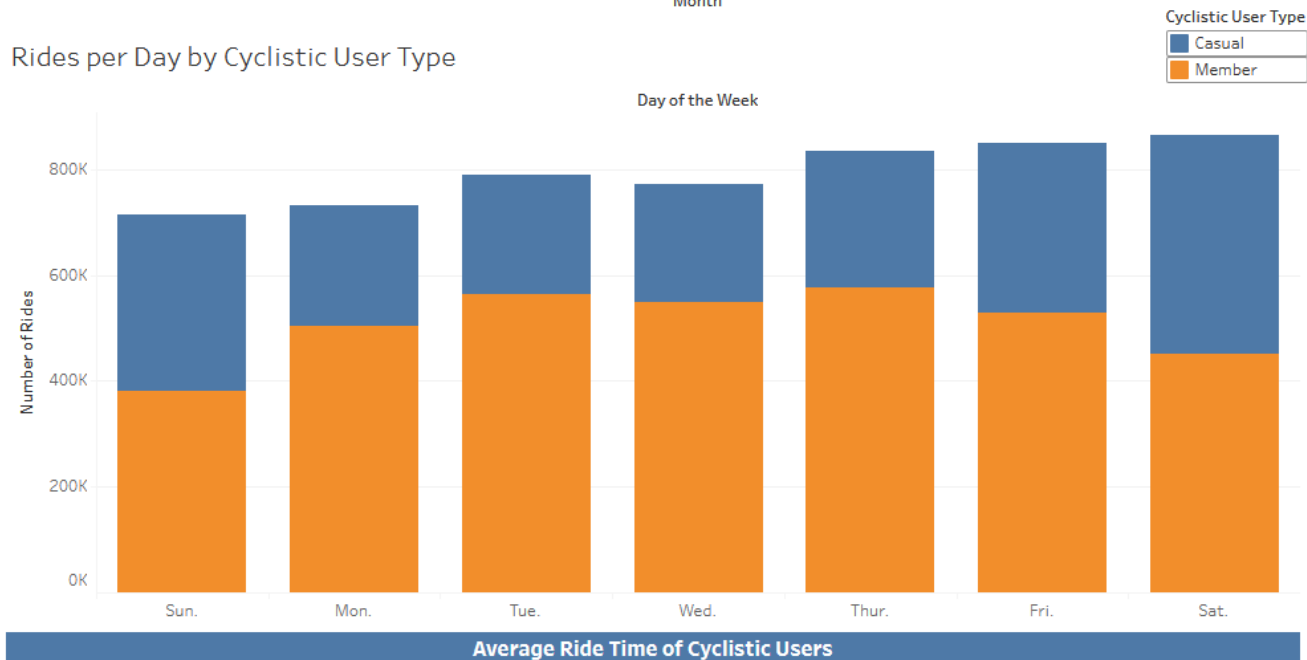
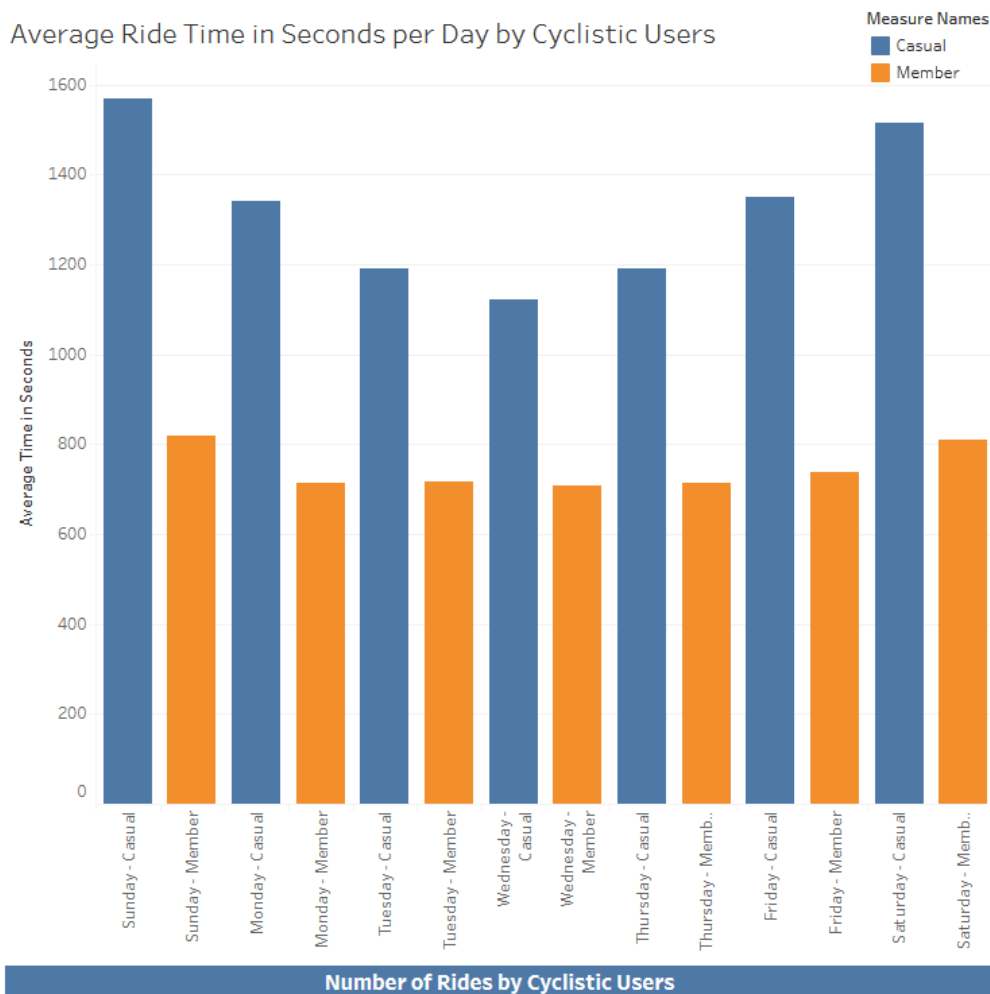


Figure 1. Monthly ridership and rides by day of week, separated by membership type.

Average Ride Time in Seconds per Day by Cyclistic Users



Average Ride Time in Seconds by Cyclistic Users

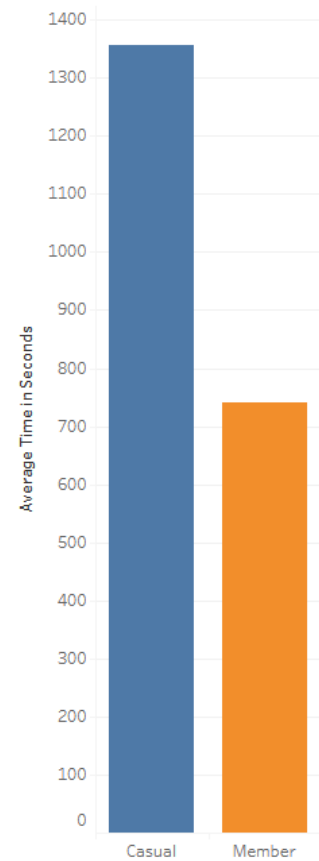


Figure 2. Average ride length by membership type and by day of week.

Interactive Tableau Public dashboard: [Cyclistic User Analysis](#)

Key findings

Casual riders take substantially longer trips: 22.6 minutes on average versus 12.3 minutes for members.

Casual usage is more weekend-oriented: Saturday is their highest-volume day, while Thursday is highest for members.

Casual usage is more seasonal: casual ridership fell 21.4% from August to September, compared with only 0.69% for members.

Annual members complete more rides in every month and use the service more consistently across the week and year.

6. Recommendations

1. Target casual riders during spring and summer, especially near peak season

Casual ridership increases sharply during the warmer months and reaches its highest level in August. Cyclistic should increase membership marketing during late spring and summer when casual riders are already actively using the service. The 21.4% decline in casual ridership from August to September suggests that late summer may be a particularly useful period for campaigns designed to retain frequent seasonal riders beyond the peak season.

2. Develop weekend-focused membership promotions

Casual riders are more concentrated on weekends, and Saturday and Sunday also produce their longest average rides. Cyclistic could target casual riders with in-app, email, or digital campaigns before and during weekends. Messaging should explain the value of membership for riders who repeatedly use the service for weekend activities rather than relying only on commuter-oriented messaging.

3. Emphasize membership value for riders taking longer, repeat leisure trips

Casual rides average 22.6 minutes, compared with 12.3 minutes for annual members, and casual ride duration is longest on weekends. Cyclistic could test membership messaging aimed at riders who use the service regularly for longer leisure trips, emphasizing the value of turning repeated single-use rides into a more predictable membership relationship. Campaign performance should be measured to determine which messages actually increase conversion.

7. Conclusion

Annual members and casual riders use Cyclistic in measurably different ways. Casual riders take much longer trips, ride more heavily on weekends, and show much stronger seasonal fluctuations. Annual members take shorter trips, complete more rides overall, and use the system more consistently throughout both the week and the year.

The strongest distinction is therefore not simply ride volume; it is the pattern of use. Casual riders appear to treat Cyclistic more like an occasional, seasonal, or recreational service, while annual members appear to incorporate the service more regularly into their transportation habits.

These differences give Cyclistic clear opportunities to focus conversion marketing on summer usage, weekend activity, and the value proposition for riders taking longer casual trips. The analysis supports targeted testing rather than one broad campaign for all casual riders.

8. Limitations and Next Steps

The trip data shows what happened, but it does not explain why each rider chose Cyclistic. It does not identify trip purpose, tourism status, household location, income, or whether a casual rider later purchased a membership. Because of this, recreational and commuting interpretations should remain cautious.

Future analysis could be strengthened by combining trip data with weather, tourism patterns, marketing campaign performance, membership conversion data, and more detailed location analysis. A next step would be to identify high-frequency casual riders and evaluate which behavioral patterns are most strongly associated with eventual membership conversion, if privacy-safe customer-level data were available.

References

Divvy / Motivate International Inc. Public trip data license: [Divvy Data License Agreement](#)

Tableau Public dashboard: [Cyclistic User Analysis](#)